Start. vv. 1–12 intact and fixed; vv. 13–29 from a starting text. Sixteen chains start from Eric Cullhed's eight reconstructions (two per start, variants of one text: 9 % of restored words differ); twenty-four more from *unrelated* starts, late states of earlier chains picked greedily so that every pair differs in 89–100 % of its restored words (`spread_starts.py`). 10 000 steps per chain, 100 warm steps, burn-in 300, a state recorded every 10 steps; run as four 2 500-step segments, each continuing the last state of the one before.

Propose a site. Draw a damaged verse $v \in \{13, \dots, 29\}$ uniformly and a letter-block span in its restored part (a mixture of block widths, or the whole restored part).

next step

Stoicheia proposes. The span is masked; 16 replacements are drawn by iterative unmasking (8 steps, sampling temperature 0.65) conditioned on the whole passage, the papyrus prefix of v kept.

Iota adscript on the letter level (the papyrus and Stoicheia); converted to subscript for the style space, the judge and the lemmata.

Formal filters (hard, in this order). Papyrus prefix kept · 26–34 letters · every word real Greek (lexicon + letter model) · **elision rule**: a remnant (δ' τ' γ' θ' τα῾ τ' …) only before a vowel, tenuis never before a rough breathing, derived aspirate only before one · particle-pair blacklist · tragic register above the Sophocles 5th percentile · no content word or lemma repeated within ±2 verses · scans as an iambic trimeter without hiatus · distinct from the current line.

Typically a few of the 16 proposals survive the filters. The 235B judge accepted 11 % of the judged proposals (10 % of all steps); the 72B 9 %; Claude Sonnet 5.5 at low effort 46 %, nearly indifferent.

any survivor?

no: reject

yes

Plain walk: $1/T_{\text{judge}} = 0$ (judge consulted and logged only). Tilted chains ($1/T_{\text{formal}} = 1$) select for Sophoclean style and are not an imputation source.

Rule set v3e for every set here (v3c + elision rule, v3d + particle term, v3e + repetition term). Earlier rule sets let the walk exploit a lax elided surface and are dropped from the dataset.

One candidate is drawn uniformly from the survivors (Metropolis-within-Gibbs: the target is Stoicheia's own distribution under the constraints).

Optional economies for a priced judge (the Claude pilot only, off for the Qwen sets): proposals changing under 10 % of a verse's letters are dropped unjudged; burn-in ends when 50 % of the restored letters differ from the start; a state is recorded when 10 % have changed since the last one.

Formal term.

$$\Delta_{\text{formal}} = \frac{1}{T_{\text{formal}}}\,\Delta\log\text{-odds(Sophocles)} + \Delta_{\text{finite}} + \Delta_{\text{particles}} + \Delta_{\text{repetition}}$$

with $1/T_{\text{formal}} = 0$: style-blind, so the walk remains a valid imputation source. The three calibration terms move the restorer's own rates to Sophocles': $\operatorname{logit} p - \operatorname{logit} q$ per verse with a finite verb (+0.02), per particle (−0.53), and $\log(p_b/q_b)$ per word by the band b of its nearest earlier occurrence (same line +0.02, 1 back +0.12, 2 back +0.24, 3–4 back −0.40, 5–7 back −0.10; weight $\frac{1}{2}$).

Mixing: after 10 000 steps the chains' final states differ in 96 % of their restored words whether or not they shared a start, yet 22–28 of 34 topics have $\hat{R} > 1.1$; the pooled lift averages over chains that have not met.

Coherence judge (Qwen3-235B-A22B-Instruct-2507, FP8, 22B active parameters per token, split over the four GH200 GPUs of one node). Window of ±2 verses on the elided surface (elision marks restored, no punctuation), old versus new line as versions A and B: *"in which version is line v the more coherent continuation, in grammar, syntax and meaning?"* Asked in both orders:

$\Delta_{\text{judge}} = \frac{1}{2}\,(\ell_{\text{new first}} - \ell_{\text{old first}})$, ℓ the next-token log-odds of answering A,

i.e. the order-averaged log-odds for the new line.

After the walk: from states to documents. Every recorded state is lemmatised by the morphological parser; the lemmata are filtered to the 3 128-lemma content vocabulary and reduced to a *bag of topic-bearing lemmata* (34 word2vec clusters, iota subscript normalised). One state is one document, about 30 topic lemmata; the sets go to the dataset `Urdatorn/bsda-miniproject-2026` with their chain set, chain and step.

Accept with probability

$\min\{1,\ \exp(\Delta_{\text{formal}} + \Delta_{\text{judge}}/T_{\text{judge}})\}$

with $1/T_{\text{judge}} = 4$ for every set in the report.

Bayesian models. *Hard*: Dirichlet–multinomial per document with Sophocles' profile as a one-lemma prior, exact posterior; the imputations' mean profile against windows of seventeen Sophoclean trimeters (and the other tragedians, Classical Attic), lift with a 90 % interval from resampling chains and windows. *Soft*: LDA with the topic–word matrix fixed, per-document NUTS. *Checks*: drift of the lift in 1 000-step windows, Gelman–Rubin $\hat{R}$ across chains, the sets from Eric's starts against the sets from unrelated starts, and the two Qwen judges against a Claude Sonnet 5.5 pilot.

accepted?

no: keep the old line

yes

Update verse v, its lemmas, finite flag, particle count, repetition bands and register; log the step with Δ_{judge}.

next step

after the last step